

The Near-Linear Boundary of Exponential Feedback

A self-consistent Lambert law, coefficient large deviations,
and a cascade of Borel-growth resonances

Research study prepared for Vladimir Reshetnikov's ProveIt program

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Abstract

Consider the uniquely defined formal solution $\widehat{U}(q) = \sum_{j \geq 1} c_j q^j \exp(\lambda_j \widehat{U}(q))$. The ProveIt feedback article leaves open the sharp growth law at the near-linear boundary $\lambda_j = jL(j)$, where L is slowly varying and unbounded: the solution diverges, although it belongs to every positive Gevrey class. Under explicit smoothness, convexity, and positive-amplitude hypotheses, we resolve this coefficient-growth problem. If $a = c_1$ and b_n is determined by

$$b_n e^{b_n} = L\left(\frac{n}{1+b_n}\right),$$

then $u_n^{1/n} \sim a e^{b_n}$. The Lagrange composition length satisfies a large-deviation principle with rate $r - 1 - \log r$ after scaling its excess by $n/(1+b_n)$. Exponentially controlled amplitudes and bounded linear perturbations of the slopes preserve these laws.

For every $\sigma > 0$, the factorial-normalized transform $\mathcal{B}_\sigma(t) = \sum u_n t^n / (n!)^\sigma$ is entire, and $\log \mathcal{B}_\sigma(t) \sim \sigma \nu_\sigma(t)$, where $\nu_\sigma^\sigma / (e^{b_{\nu_\sigma}}) = at$. This identifies a positive-direction obstruction to every finite-order Borel–Laplace summation procedure of the standard Gevrey kind. For $L(j) = \exp((\log j)^\gamma)$, $0 < \gamma < 1$, an explicit power–logarithmic reversion gives all nonvanishing terms in the logarithm of the growth normalizer. At $\gamma = 1 - 1/m$, a term becomes constant and produces the factor $\exp(1/m)$ in ordinary Borel growth. A separate moving-argument example shows why replacing the implicit Lambert law by $W(L(n))$ can lose a constant or an unbounded correction. Complete proofs, finite coefficient bounds, reproducible exact checks, and ten further research directions are included.

Scope and status. The main results are proved here in ordinary mathematics. They answer a stated repository question under the hypotheses specified below; they are not a claim to settle an unrestricted problem in transseries theory. No new Lean proof is supplied. Classical inversion and regular-variation tools are credited, and publication-level priority has not been established by an exhaustive literature search.

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1 The repository question and the contribution

The relevant ProveIt material now lives under **Analysis/Transseries/**. Its main volume develops power–logarithmic calculus, formal reversion, exact cores, residual transport, and examples of special-function inversion. The separate feedback article [1] studies the particularly transparent countable kernel

$$\widehat{U}(q) = \sum_{j \geq 1} q^j \exp(\lambda_j \widehat{U}(q)). \quad (1.1)$$

It supplies a finite positive coefficient formula, a sharp convergence criterion, a Gevrey classification, and logarithmic coefficient asymptotics for regularly varying slopes of index strictly greater than one. Its section entitled *The near-linear transition* explicitly asks for sharp coefficient asymptotics and natural summation scales when $\lambda_j = j(\log j)^\beta$ or, more generally, $jL(j)$ with unbounded slowly varying L . The difficulty is not formal existence. It is the change of scale when the positive multiple of $n \log n$ in the superlinear analysis disappears.

This article treats that question directly. It also answers a restricted but useful part of the same source’s amplitude question. We allow

$$\widehat{U}(q) = \sum_{j \geq 1} c_j q^j \exp(\lambda_j \widehat{U}(q)), \quad c_j > 0, \quad \lambda_j \geq 0. \quad (1.2)$$

The central object is not $W(L(n))$ but a Lambert equation with a *moving argument*. This distinction is necessary at the precision of an equivalent for $u_n^{1/n}$, rather than merely an equivalent for $\log u_n$.

1.1 What is established

The main coefficient theorem gives

$$\log u_n = n(\log a + b_n + o(1)), \quad a = c_1. \quad (1.3)$$

It is accompanied by a complete large-deviation law for the number of primitive factors in the finite Lagrange expansion. A second theorem transfers this precise coefficient scale to the growth of all factorial-normalized entire transforms. A third theorem computes the resulting Borel-growth transseries for stretched-logarithmic slopes, including its constant terms at every transition value $1 - 1/m$.

These are stronger than a determination of the least Gevrey index, but we distinguish them carefully from a multiplicative equivalent for u_n itself. Equation (1.3) does *not* assert $u_n \sim (ae^{b_n})^n$: an $\exp(o(n))$ factor remains unresolved. Similarly, the large-deviation result is not a central limit theorem, and a growth-adapted integral is not automatically an acceleration operator compatible with the original equation.

1.2 Relation to established tools and novelty

The finite coefficient identity is classical Lagrange inversion [2], already specialized to unit amplitudes in the repository. General inversion with infinitely many colours has a much broader literature; Jansen, Kuna, and Tsagkarogiannis [3] are a relevant reference, not a source for the particular asymptotic laws proved here. Slow variation and its use in inversion are standard themes of regular-variation theory [4]. We use a differentiable subclass so that the needed estimates can be proved directly rather than imported as a large technical package.

The new work relative to the inspected repository is the borderline Lambert normalizer, its probability law, its positive-amplitude and slope-perturbation stability, and its consequences for Borel growth and moving power-logarithmic terms. The coefficients in the final scalar reversion are classical Lagrange coefficients; we do not claim their formula as a new inversion theorem. Nor do we claim an exhaustive global priority result for the combined conclusions. The exact source snapshot and claim boundaries are recorded in Appendix A.

2 Hypotheses, normalizers, and theorem statements

Definition 2.1 (Admissible near-linear slopes). An admissible slope profile is a positive, nondecreasing C^1 function $L : [1, \infty) \rightarrow (0, \infty)$ such that

$$L(x) \longrightarrow \infty, \quad \varepsilon(x) := \frac{xL'(x)}{L(x)} \longrightarrow 0, \quad (2.1)$$

and $\Lambda(x) = xL(x)$ is convex. The base slopes are $\lambda_j = \Lambda(j)$. More generally, the perturbation theorem allows nonnegative $\tilde{\lambda}_j$ with

$$|\tilde{\lambda}_j - \Lambda(j)| \leq Cj \quad (j \geq 1) \quad (2.2)$$

for a fixed C . Conditions imposed only for large x may be used after choosing an admissible extension; finite discrepancies are covered by (2.2).

Condition (2.1) implies slow variation: integrate L'/L between x and rx to obtain $L(rx)/L(x) \rightarrow 1$ for each fixed $r > 0$. It also gives $\log L(x) = o(\log x)$.

Definition 2.2 (Exponentially controlled positive amplitudes). Put $a = c_1 > 0$ and $d_j = c_j/a^j$. We require a constant $D \geq 0$ such that

$$e^{-D(j-1)} \leq d_j \leq e^{D(j-1)} \quad (j \geq 1). \quad (2.3)$$

Thus $d_1 = 1$. Polynomial amplitudes, positive constant amplitudes, positive geometric amplitudes, and finite modifications of such families are included. Superexponentially small amplitudes and missing actions are not included.

For all sufficiently large real x , define $b(x) > 0$ by

$$b(x)e^{b(x)} = L\left(\frac{x}{1+b(x)}\right), \quad f(x) = e^{b(x)}, \quad x_*(x) = \frac{x}{1+b(x)}. \quad (2.4)$$

We abbreviate $b_n = b(n)$ and $x_n = x_*(n)$. Existence, uniqueness, and regularity are proved in Section 4.

Theorem 2.3 (Sharp borderline coefficient law). *Under Definitions 2.1 and 2.2, including the perturbation option (2.2), the unique formal solution $\hat{U} = \sum_{n \geq 1} u_n q^n$ has positive coefficients and satisfies*

$$\boxed{u_n^{1/n} \sim af(n) = ae^{b_n}.} \quad (2.5)$$

In particular, its radius of convergence is zero, while for every $\sigma > 0$ and every $A > 0$ there is a finite $C_{\sigma,A}$ such that

$$u_n \leq C_{\sigma,A} A^n (n!)^\sigma \quad (n \geq 1). \quad (2.6)$$

The arbitrary $A > 0$ in (2.6) says more than positive Gevrey membership: the factorial-normalized type is zero at every positive order. The series still fails to be convergent.

For the probabilistic refinement, set

$$W_{n,k} = \frac{1}{k!} \sum_{\substack{j_1 + \dots + j_k = n \\ j_i \geq 1}} \left(\prod_{i=1}^k c_{j_i} \right) \left(\sum_{i=1}^k \lambda_{j_i} \right)^{k-1}. \quad (2.7)$$

As usual, a zero exponent is interpreted as one, including 0^0 here. Then $u_n = \sum_k W_{n,k}$. Give K_n the distribution $\mathbb{P}(K_n = k) = W_{n,k}/u_n$, and define

$$R_n = \frac{n - K_n + 1}{x_n}, \quad I(r) = r - 1 - \log r \quad (r > 0). \quad (2.8)$$

Theorem 2.4 (Length large deviations). *The variables R_n obey a large-deviation principle on $(0, \infty)$ with speed n and good rate function I . Explicitly, for open G and closed F in $(0, \infty)$,*

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \log \mathbb{P}(R_n \in G) \geq - \inf_{r \in G} I(r), \quad (2.9)$$

$$\limsup_{n \rightarrow \infty} \frac{1}{n} \log \mathbb{P}(R_n \in F) \leq - \inf_{r \in F} I(r). \quad (2.10)$$

If $(n - k_n + 1)/x_n \rightarrow r \in (0, \infty)$, then the local statement is

$$\frac{1}{n} \log \frac{W_{n,k_n}}{u_n} \rightarrow -I(r). \quad (2.11)$$

Consequently $R_n \rightarrow 1$ in probability, exponentially at every fixed positive distance from one.

Theorem 2.5 (Factorial-Borel growth). *For $\sigma > 0$, the function*

$$\mathcal{B}_\sigma(t) = \sum_{n \geq 1} \frac{u_n t^n}{(n!)^\sigma} \quad (2.12)$$

is entire. Let $\nu_\sigma(t)$ be the eventually unique large positive solution of

$$\frac{\nu_\sigma(t)^\sigma}{f(\nu_\sigma(t))} = at. \quad (2.13)$$

Then, on the positive real ray,

$$\boxed{\log \mathcal{B}_\sigma(t) \sim \sigma \nu_\sigma(t)}. \quad (2.14)$$

The maximum modulus equals $\mathcal{B}_\sigma(t)$ on the circle $|z| = t$. Its order is $1/\sigma$ and its type at that order is infinite.

The theorem gives an implicit asymptotic equivalent, not merely an order of growth. It is that extra precision which detects the constant factors in Section 9.

3 Formal existence and finite coefficient inequalities

Proposition 3.1 (Formal locality and coefficient formula). *Equation (1.2) has a unique solution in $q\mathbb{R}[[q]]$. The coefficient u_n depends only on the actions and slopes with index at most n , and it is given by $u_n = \sum_{k=1}^n W_{n,k}$.*

Proof. The map $V \mapsto \sum_j c_j q^j \exp(\lambda_j V)$ raises the q -adic valuation of a difference by at least one. Starting at zero therefore stabilizes every coefficient and gives a unique fixed point. In degree n , only $j \leq n$ and finitely many exponential terms enter.

For the formula, introduce a marker z and solve $V = z \sum_j c_j q^j \exp(\lambda_j V)$. Formal Lagrange inversion gives

$$[z^k]V = \frac{1}{k} [t^{k-1}] \left(\sum_j c_j q^j \exp(\lambda_j t) \right)^k.$$

The ordered composition $(j_1, \dots, j_k)$ contributes the coefficient of t^{k-1} in $\exp((\sum_i \lambda_{j_i})t)$, proving (2.7). Specializing $z = 1$ is coefficientwise legitimate because $k \leq n$ in degree q^n . Positivity follows from the $k = 1$ term $c_n > 0$. $\square$

For example, without any asymptotic hypotheses,

$$\begin{aligned} u_1 &= c_1, \\ u_2 &= c_2 + c_1^2 \lambda_1, \\ u_3 &= c_3 + c_1 c_2 (\lambda_1 + \lambda_2) + \frac{3}{2} c_1^3 \lambda_1^2. \end{aligned}$$

The k -marker will be important: a formal coefficient formula does not by itself say which composition lengths supply its mass.

Lemma 3.2 (Convex composition bound). *If the base sequence $\lambda_j = \Lambda(j)$ is convex, then for $j_1 + \dots + j_k = n$,*

$$\sum_{i=1}^k \lambda_{j_i} \leq \lambda_{n-k+1} + (k-1)\lambda_1. \quad (3.1)$$

Proof. The forward differences $\lambda_{j+1} - \lambda_j$ are nondecreasing. If $r \geq s > 1$, moving one unit from s to r cannot decrease the sum $\lambda_r + \lambda_s$. Repeating the move leaves at most one part larger than one. Its size is $n - k + 1$, which proves the inequality. $\square$

Write $m = n - k + 1$ and

$$A_{n,k} = \lambda_m + (k-1)\lambda_1, \quad E_{n,k} = \frac{A_{n,k}^{k-1}}{k!}, \quad r_{n,k} = \begin{cases} k, & m > 1, \\ 1, & m = 1. \end{cases} \quad (3.2)$$

For unit amplitudes, one composition with a single part m and the remaining parts one has slope sum $A_{n,k}$. It has k distinct placements unless $m = 1$.

Proposition 3.3 (Finite, computable coefficient bounds). *For the unperturbed convex slopes and amplitudes (2.3),*

$$a^n e^{-D(n-k)} r_{n,k} E_{n,k} \leq W_{n,k} \leq a^n e^{D(n-k)} \binom{n-1}{k-1} E_{n,k}. \quad (3.3)$$

In particular,

$$\max_k a^n e^{-D(n-k)} r_{n,k} E_{n,k} \leq u_n \leq \sum_k a^n e^{D(n-k)} \binom{n-1}{k-1} E_{n,k}. \quad (3.4)$$

Proof. Each composition satisfies

$$\prod_i c_{j_i} = a^n \prod_i d_{j_i}, \quad e^{-D(n-k)} \leq \prod_i d_{j_i} \leq e^{D(n-k)},$$

because $\sum_i (j_i - 1) = n - k$. Use the exceptional-part compositions for the lower bound, and (3.1) and the exact number $\binom{n-1}{k-1}$ of compositions for the upper bound. $\square$

These bounds require only $O(n)$ envelope evaluations once the first n slopes are known. They can be evaluated with directed rounding to obtain numerical intervals. The supplied floating-point tables do not implement that rounding; the small rational tests check the inequalities exactly.

4 The self-consistent Lambert normalizer

Lemma 4.1 (Existence and regularity). *For large x , (2.4) has exactly one solution with $0 < b(x) < x - 1$. Moreover,*

$$b(x) \rightarrow \infty, \quad b(x) = o(\log x), \quad x_*(x) \rightarrow \infty, \quad (4.1)$$

$$b(x) \sim \log L(x), \quad \frac{db(x)}{d \log x} = \frac{\varepsilon(x_*(x))}{1 + 1/b(x) + \varepsilon(x_*(x))/(1 + b(x))} \rightarrow 0. \quad (4.2)$$

Thus $f(x) = e^{b(x)}$ is nondecreasing, unbounded, and slowly varying. For each $\sigma > 0$, $x^\sigma/f(x)$ is eventually strictly increasing, unbounded, and regularly varying with index σ .

Proof. As a function of b , the left side be^b is strictly increasing and the right side $L(x/(1+b))$ is nonincreasing. At $b = 0$ the former is zero; at $b = x - 1$ it exceeds $L(1)$ for large x . This proves existence and uniqueness. Since $b \leq W(L(x)) \leq \log L(x)$ once $b \geq 1$, we have $b = o(\log x)$, so $x_*(x) \rightarrow \infty$. If b remained bounded along a subsequence, the defining equation would contradict $L(x_*) \rightarrow \infty$.

For large x , $0 \leq \varepsilon \leq 1$ throughout $[x_*, x]$. Integration gives

$$0 \leq \log L(x) - \log L(x_*) \leq \log(1 + b).$$

But $\log L(x_*) = b + \log b$, so $\log L(x) = b + O(\log b)$, proving the claimed equivalent. Implicit differentiation of $b + \log b = \log L(x/(1+b))$ gives (4.2). Integration of that derivative on fixed logarithmic intervals proves slow variation of f . Finally,

$$\frac{d}{d \log x} \log \frac{x^\sigma}{f(x)} = \sigma - \frac{db}{d \log x} \rightarrow \sigma > 0,$$

which proves the remaining assertions. $\square$

The same calculation shows that ν_σ in (2.13) exists uniquely for large arguments. Its regular variation index is $1/\sigma$: taking ratios in the defining equation and using the derivative bound gives this directly.

4.1 Why the argument moves

Ignoring the fixed λ_1 contribution for the moment, the exceptional- part term with m large has logarithm

$$(n - m) \left(\log \frac{mL(m)}{n - m} + 1 \right) + O(\log n).$$

Write $y = (n - m)/m$. Its logarithm divided by n becomes

$$\frac{y}{1 + y} \left(\log L \left(\frac{n}{1 + y} \right) - \log y + 1 \right). \quad (4.3)$$

If $y = b_n$, the expression equals *exactly* b_n . This cancellation is the reason for the normalization (2.4). The actual finite- n maximizer need not have $y = b_n$; the proof must show that optimizing it changes the normalized logarithm by only $o(1)$. We do that next, including the full composition count rather than discarding it.

5 The borderline envelope and the length law

All statements in this section first concern the unperturbed convex slopes. Put $b = b_n$, $x_* = n/(1 + b)$, $\rho = m/n$, and define

$$H_n(\rho) = (1 - \rho) \left(\log \frac{\rho L(n\rho)}{1 - \rho} + 1 \right), \quad (5.1)$$

$$H_n^{(c)}(\rho) = (1 - \rho) \left(\log \left(\frac{\rho L(n\rho)}{1 - \rho} + c \right) + 1 \right), \quad (5.2)$$

where $c = \lambda_1$. Endpoint values at $\rho = 1$ are understood by limits. At $\rho = m/n$, Stirling's formula, uniformly in $1 \leq k \leq n$, gives

$$\frac{1}{n} \log E_{n,k} = H_n^{(c)}(\rho) + o(1), \quad k = n - m + 1. \quad (5.3)$$

The harmless shifts by one contribute at most $O(\log(nL(n))/n)$. For $k = 1$ the exact left side is zero, consistent with the endpoint. Also, with $h(\rho) = -\rho \log \rho - (1 - \rho) \log(1 - \rho)$,

$$\frac{1}{n} \log \binom{n-1}{k-1} = h(\rho) + o(1) \quad (5.4)$$

uniformly. Both uniform estimates follow, for example, from the factorial bounds $\log(r!) = r \log r - r + O(\log(r + 2))$.

Lemma 5.1 (Compact profile and coercivity). *Uniformly for r in compact subsets of $(0, \infty)$,*

$$H_n \left(\frac{r}{1 + b_n} \right) - b_n \longrightarrow \log r + 1 - r. \quad (5.5)$$

For fixed $D \geq 0$ and $c \geq 0$,

$$\max_{1 \leq m \leq n} \left\{ H_n^{(c)}(m/n) + h(m/n) + D \frac{m}{n} \right\} = b_n + o(1). \quad (5.6)$$

Moreover the expression minus b_n is coercive in $r = m/x_n$: along any sequence with $r \rightarrow 0$ or $r \rightarrow \infty$, it tends to $-\infty$.

Proof. For $\rho = r/(1+b)$, let $\Delta_n(r) = \log L(rx_*) - \log L(x_*)$. Since $\log L(x_*) = b + \log b$, a direct calculation gives

$$H_n(\rho) - b = (1 - \rho) \left(\log \frac{b}{1+b-r} + \log r + \Delta_n(r) + 1 \right) - b\rho. \quad (5.7)$$

On a fixed compact r -interval, $\Delta_n(r) \rightarrow 0$ uniformly by (2.1), $\rho \rightarrow 0$, and $b\rho \rightarrow r$. This proves (5.5).

We give the tail argument explicitly, since a bound by 2^n throughout would leave an erroneous constant of order n . First, if $\rho \geq \eta > 0$, then

$$H_n(\rho) \leq (1 - \eta) \log L(n) + O(1) \leq (1 - \eta)b + O(\log b).$$

Here $(1 - \rho) \log(\rho/(1 - \rho))$ is bounded above on $(0, 1)$. The entropy and $D\rho$ are bounded, so these terms are far below b .

The addition of c causes no problem. Put $X = \rho L(n\rho)/(1 - \rho)$. If $X \geq e^{b/2}$, then

$$0 \leq H_n^{(c)} - H_n \leq \log(1 + ce^{-b/2}) = o(1).$$

If $X < e^{b/2}$, then $H_n^{(c)} \leq b/2 + O(1)$, which is again far below b even after entropy and amplitude terms. Therefore every potentially competitive sequence must have $\rho \rightarrow 0$ and may be analyzed using H_n ; then $h(\rho) + D\rho \rightarrow 0$.

Suppose now that $\rho \rightarrow 0$ but $r \rightarrow \infty$. Both x_* and rx_* are large. Given $\delta > 0$, integration of (2.1) gives $0 \leq \Delta_n(r) \leq \delta \log r$ eventually. In (5.7), $\log(b/(1+b-r)) = o(1)$, while $b\rho = r(1 + o(1))$. Thus its right side is bounded above by

$$(1 + \delta) \log r + 1 - r(1 + o(1)) + o(1),$$

which tends to $-\infty$. If $r \rightarrow 0$ and $m \rightarrow \infty$, monotonicity gives $\Delta_n(r) \leq 0$; the $\log r$ term then forces the same conclusion. Bounded m is even simpler: $H_n^{(c)}$ stays bounded above, or tends downward, while $b \rightarrow \infty$.

These arguments prove coercivity. Any sequence of maximizers is therefore confined to a compact r -interval after discarding a vanishing error. On such an interval, the limiting profile is $\log r + 1 - r$, whose unique maximum is zero at $r = 1$. Taking m nearest to x_n gives the matching lower bound. This proves (5.6). $\square$

Proof of Theorems 2.3 and 2.4 for the base slopes. Use (3.4). The upper logarithm differs from the maximum of its n summands by at most $\log n = o(n)$. Equations (5.3)–(5.6) show that this maximum divided by n is $\log a + b_n + o(1)$. For the lower bound choose m nearest to x_n . Then $D(n - k)/n \rightarrow 0$, and the compact profile at $r = 1$ gives the same value. This proves (2.5).

For $m/x_n \rightarrow r \in (0, \infty)$, the upper and lower estimates for $W_{n,k}$ differ after division by n by at most $2D(m - 1)/n + h(m/n) + o(1) = o(1)$. Hence, uniformly on compact r -intervals,

$$\frac{1}{n} \log W_{n,k} - \log a - b_n \longrightarrow -I(r). \quad (5.8)$$

Subtracting the coefficient law proves (2.11).

For an open set, choose one lattice point m/x_n approaching any of its points; $x_n \rightarrow \infty$ makes this possible. This proves the lower large-deviation bound. On a compact set, the largest of at most n weights gives the upper bound, since $\log n/n \rightarrow 0$. Lemma 5.1's coercivity implies exponential tightness: for every $M > 0$ there are $0 < \delta < R < \infty$ such that the total probability

outside $[\delta, R]$ is at most $\exp(-Mn + o(n))$. This extends the upper bound to every closed set and completes the LDP. Since I vanishes only at one, exponential concentration follows.

Finally $b_n \rightarrow \infty$ proves radius zero, and $b_n = o(\log n)$ combined with Stirling's formula proves (2.6), including arbitrary $A > 0$. $\square$

Remark 5.2 (What length concentration does not say). The variable $n - K_n$ is the total excess $\sum_i (j_i - 1)$, not the largest part. A composition with one exceptional action is enough to prove the lower envelope, but the LDP does not by itself prove that a typical weighted composition has a unique giant action. Near index one, the penalty for splitting a large action can be smaller than order n . Identifying that smaller scale is a separate problem.

6 Robustness under linear slope errors

Proposition 6.1 (Stability at root and large-deviation precision). *The coefficient law and the length LDP remain valid when the slopes are replaced by any nonnegative sequence satisfying (2.2). The same normalizer b_n is used.*

Proof. For a composition of n , let $S = \sum_i \Lambda(j_i)$ and $\tilde{S} = \sum_i \tilde{\lambda}_{j_i}$. Then $|\tilde{S} - S| \leq Cn$, independently of its length. Split the finite sum into high terms with $S \geq ne^{b_n/2}$ and low terms with $S < ne^{b_n/2}$.

On the high set,

$$\frac{\tilde{S}}{S} = 1 + O(e^{-b_n/2}), \quad \left| \log \frac{\tilde{S}^{k-1}}{S^{k-1}} \right| \leq O(ne^{-b_n/2}) = o(n)$$

uniformly. The amplitudes are unchanged. On the low set, both slope sums are at most $n(e^{b_n/2} + C)$. Because this bound divided by n tends to infinity, $A^{k-1}/k!$ with $A = n(e^{b_n/2} + C)$ increases for $1 \leq k \leq n$, up to an immaterial first factor. Its maximum has logarithm $n(b_n/2 + 1 + o(1))$. There are at most 2^{n-1} compositions altogether, and their amplitude product is at most $a^n e^{Dn}$. The total low contribution therefore has logarithm at most

$$n(\log a + b_n/2 + D + 1 + \log 2 + o(1)),$$

which is negligible relative to $n(\log a + b_n)$.

Thus the two total coefficient logarithms differ by $o(n)$. At every fixed scaled length $r > 0$, the exceptional-part lower bound has S/n of order e^{b_n} , so it is on the high set. The local LDP and its compact upper bounds therefore transfer with error $o(n)$. The same high/low split preserves exponential tightness, proving the full LDP for the perturbed slopes. $\square$

This permits convenient exact models and harmless changes at small indices. It does not license a replacement of $L(x_n)$ by $L(n)$: that would change each slope at large indices by more than a fixed linear error in some of our examples.

7 From coefficient growth to entire Borel growth

We prove a general maximum-term lemma, making explicit the input needed from the coefficient theorem.

Lemma 7.1 (Slowly varying coefficient transfer). *Let $a > 0$, let f be positive, nondecreasing, unbounded, and C^1 for large arguments, and assume $d \log f(x)/d \log x \rightarrow 0$. If $v_n > 0$ satisfy $v_n^{1/n} \sim af(n)$, then for $\sigma > 0$,*

$$\log \left(\sum_{n \geq 1} \frac{v_n t^n}{(n!)^\sigma} \right) \sim \sigma N(t), \quad N(t)^\sigma / f(N(t)) = at.$$

Proof. Since $f(n) = n^{o(1)}$, the series is entire. Write $\log v_n = n(\log a + \log f(n) + \eta_n)$, where $\eta_n \rightarrow 0$, and put $N = N(t)$. Stirling's formula gives

$$\log \frac{v_n t^n}{(n!)^\sigma} = n \left(\sigma \log \frac{N}{n} + \log \frac{f(n)}{f(N)} + \sigma + \eta_n \right) + O(\log(n+2)). \quad (7.1)$$

For $n/N \rightarrow r$ in a compact subinterval of $(0, \infty)$, divide by N : the right side tends uniformly to $\sigma r(1 - \log r)$. Its unique maximum is σ at $r = 1$. The nearest integer to N therefore supplies the lower bound.

For completeness, control the tails. For $n \leq \delta N$, beyond a fixed initial segment, monotonicity gives $f(n) \leq f(N)$, and $|\eta_n|$ is small. The upper logarithm divided by N is bounded by $\delta(\sigma \log(1/\delta) + \sigma + o(1))$ when δ is small. The fixed initial segment has logarithm $O(\log t) = o(N)$.

For $n \geq RN$, the derivative hypothesis gives, uniformly, $f(n)/f(N) \leq (n/N)^{\sigma/2}$ once N is large. Formula (7.1) is then at most

$$n \left(-\frac{\sigma}{2} \log(n/N) + \sigma + o(1) \right).$$

For sufficiently large fixed R , these terms have a summable geometric upper bound, and their total is exponentially below the central maximum. On $[\delta N, RN]$ there are $O(N)$ terms, so summing adds only $O(\log N) = o(N)$ to the maximum logarithm. Letting the compact interval expand proves the result. $\square$

Proof of Theorem 2.5. Apply Lemma 7.1 with the coefficient theorem and Lemma 4.1. Nonnegative coefficients give $|\mathcal{B}_\sigma(z)| \leq \mathcal{B}_\sigma(|z|)$, with equality on the positive ray. The regular variation of ν_σ has index $1/\sigma$, and

$$\frac{\nu_\sigma(t)}{t^{1/\sigma}} = (af(\nu_\sigma(t)))^{1/\sigma} \rightarrow \infty.$$

These facts prove the order and infinite-type assertions. $\square$

7.1 Standard Gamma normalization and summability

For clarity, the standard moment normalization associated with Gevrey order σ is

$$\mathfrak{B}_\sigma(t) = \sum_{n \geq 1} \frac{u_n t^n}{\Gamma(1 + \sigma n)}. \quad (7.2)$$

Our $\mathcal{B}_1$ is an unshifted Borel transform; differentiating it gives the common shifted convention. The obstruction below is independent of this harmless convention.

Corollary 7.2 (No positive-direction standard summation of finite order). *For every $\sigma > 0$, $\mathfrak{B}_\sigma$ is entire and*

$$\log \mathfrak{B}_\sigma(t) \sim \sigma \nu_\sigma(t/\sigma^\sigma), \quad t \rightarrow +\infty. \quad (7.3)$$

In particular, for every $A > 0$,

$$\int_1^\infty t^{1/\sigma-1} \exp(-At^{1/\sigma}) \mathfrak{B}_\sigma(t) dt = +\infty. \quad (7.4)$$

Thus the formal solution is not summable in the positive direction by any standard finite-order Gevrey Borel–Laplace procedure.

Proof. Stirling’s formula gives

$$\log \Gamma(1 + \sigma n) - \sigma \log(n!) = \sigma n \log \sigma + O(\log(n + 2)).$$

Consequently the maximum-term proof applies with t replaced by t/σ^σ ; the polynomial factors do not affect the leading logarithm. The quotient of (7.3) by $t^{1/\sigma}$ tends to infinity. The integrand in (7.4) therefore eventually exceeds an exponentially increasing function, which proves divergence. $\square$

Borel–Laplace summability requires both continuation and growth control; see Sauzin [5]. The transforms here already continue to the entire plane. Their obstruction is growth on the chosen ray, not a finite singularity. Gevrey implicit-function theorems such as Nikolaev’s [6] have analytic hypotheses which cannot be replaced by coefficient membership alone. Nothing in Corollary 7.2 excludes useful sums in other directions.

Proposition 7.3 (A critical positive damping scale). *For the ordinary Borel transform, put $\nu = \nu_1$. Then*

$$\int_1^\infty \exp(-A\nu(t)) \mathcal{B}_1(t) dt$$

converges for $A > 1$ and diverges for $0 < A < 1$.

Proof. Use $\log \mathcal{B}_1(t) = (1 + o(1))\nu(t)$ and $\nu(t)/t = af(\nu(t)) \rightarrow \infty$. $\square$

This is a growth statement only. It does not define an acceleration operator respecting products, differentiation, or the feedback equation. The borderline $A = 1$ also requires information not provided by the leading equivalent.

7.2 A literal positive fixed point is impossible

Under our amplitude condition and $L(j) \rightarrow \infty$, for $q > 0$ and $u > 0$

$$\log(c_j q^j e^{\lambda_j u}) \geq j(\log a + \log q - D + uL(j)) + O(1) \longrightarrow +\infty$$

for base slopes, and the same is true after a bounded linear slope perturbation. Thus the summands do not tend to zero. No proposed positive-ray regularization can be described as a solution of the literal convergent positive sum without changing its analytic interpretation.

8 Explicit profiles and the moving-argument correction

The implicit normalizer is already an asymptotic equivalent. To extract its visible power–logarithmic terms, write

$$s = \log x, \quad h(s) = \log L(e^s). \quad (8.1)$$

The notation h here is unrelated to the binary entropy used in Section 5.

Proposition 8.1 (A moving-argument expansion). *Suppose the admissible profile has $h \in C^2$ eventually and*

$$(\log h(s))^2 \sup_{|v-s| \leq 2 \log h(s)} |h''(v)| \longrightarrow 0. \quad (8.2)$$

Then

$$b(e^s) = h(s) - \log h(s) - h'(s) \log h(s) + o(1). \quad (8.3)$$

In particular,

$$b(e^s) - W(L(e^s)) = -h'(s) \log h(s) + o(1). \quad (8.4)$$

Proof. Lemma 4.1 gives $b \sim h(s)$, so $\log b = \log h(s) + o(1)$ and $\log(1+b) = \log h(s) + o(1)$. Also $h'(s) = \varepsilon(e^s) \rightarrow 0$. In the equation

$$b + \log b = h(s - \log(1+b)),$$

we may therefore replace the right-hand argument by $s - \log h(s)$ at an additive cost $o(1)$. Taylor's theorem and (8.2) give

$$h(s - \log h(s)) = h(s) - h'(s) \log h(s) + o(1),$$

which proves (8.3). Finally the positive solution of $w + \log w = h$ satisfies $w = h - \log h + o(1)$: substituting the proposed value leaves an $o(1)$ residual, and the derivative $1 + 1/w$ is bounded below by one. This proves (8.4); the conventional Lambert notation agrees with DLMF §4.13 [7]. $\square$

8.1 Logarithmic slope amplification

Let $\beta > 0$ and suppose $L(x) = (\log x)^\beta$ for large x , with an admissible extension at small x . Then $h(s) = \beta \log s$, and the derivative correction in (8.3) tends to zero. Consequently

$$b_n = \beta \log \log n - \log \log \log n - \log \beta + o(1), \quad (8.5)$$

$$u_n^{1/n} \sim a \frac{(\log n)^\beta}{\beta \log \log n}. \quad (8.6)$$

This resolves the explicit $j(\log j)^\beta$ family in the repository's near-linear question at coefficient-root precision.

For the ordinary Borel transform,

$$\log \mathcal{B}_1(t) \sim at \frac{(\log t)^\beta}{\beta \log \log t}. \quad (8.7)$$

Indeed, $f(x) \sim (\log x)^\beta / (\beta \log \log x)$ satisfies $f(atf(t))/f(t) \rightarrow 1$, so the inverse relation $\nu_1 = atf(\nu_1)$ gives $\nu_1 \sim atf(t)$. This last conclusion can also be proved directly by substituting $(1 \pm \eta)atf(t)$ and using monotonicity, then letting $\eta \downarrow 0$.

More generally, putting $R = \sigma^{-1} \log(at)$ yields

$$\log \mathcal{B}_\sigma(t) \sim \sigma (at)^{1/\sigma} \left(\frac{R^\beta}{\beta \log R} \right)^{1/\sigma}. \quad (8.8)$$

All constants, including the σ inside R , matter in this formula.

8.2 An exact rational example

The slopes

$$\lambda_j = jH_j, \quad H_j = \sum_{r=1}^j \frac{1}{r}, \quad (8.9)$$

are rational and convex: their successive difference is $H_j + 1$, which increases. They differ from $j \log j$ by $O(j)$, so Proposition 6.1 applies. With unit amplitudes,

$$u_n^{1/n} \sim \frac{\log n}{\log \log n}, \quad \log \mathcal{B}_1(t) \sim \frac{t \log t}{\log \log t}. \quad (8.10)$$

For a concrete smooth admissible interpolation used in the numerical envelopes, set $L(x) = \psi(x+1) + \gamma_E$. The identity

$$L(x) = \sum_{k \geq 1} \left(\frac{1}{k} - \frac{1}{x+k} \right)$$

implies $L' > 0$ and

$$(xL(x))'' = 2 \sum_{k \geq 1} \frac{k}{(x+k)^3} > 0.$$

Also $L(x) \sim \log x$ and $xL'(x)/L(x) \rightarrow 0$, directly by integral comparison. Thus the interpolation is admissible without an appeal to any unproved numerical property of the digamma function.

8.3 Stretched-logarithmic amplification

For $0 < \gamma < 1$, let $L(x) = \exp((\log x)^\gamma)$ eventually. Here $h(s) = s^\gamma$, and Proposition 8.1 gives

$$b_n = (\log n)^\gamma - \gamma \log \log n + o(1), \quad u_n^{1/n} \sim a \frac{\exp((\log n)^\gamma)}{(\log n)^\gamma}. \quad (8.11)$$

Although the moving argument has only an $o(1)$ effect on this particular b_n expansion, substituting t for $\nu_1(t)$ in the Borel inversion need not have an $o(1)$ effect. That second inversion is the source of the cascade in Section 9.

8.4 A family where the naive Lambert approximation fails

Fix $\eta > 0$ and take

$$L(e^s) = \exp\left(\frac{s}{(\log s)^\eta}\right) \quad \text{for sufficiently large } s. \quad (8.12)$$

The admissibility and second-derivative conditions hold eventually. With $\ell = \log s$, direct differentiation gives

$$h = \frac{s}{\ell^\eta}, \quad \log h = \ell - \eta \log \ell, \quad h' \log h = \ell^{1-\eta} + o(1).$$

Therefore

$$b(e^s) = \frac{s}{\ell^\eta} - \ell + \eta \log \ell - \ell^{1-\eta} + o(1). \quad (8.13)$$

It follows that

$$\frac{e^{b(n)}}{e^{W(L(n))}} = \exp\left(-(\log \log n)^{1-\eta} + o(1)\right). \quad (8.14)$$

For $\eta > 1$ the naive replacement happens to be valid at root precision. At $\eta = 1$ it loses the factor e^{-1} , and for $0 < \eta < 1$ its relative error is unbounded. In particular, at $\eta = 1$,

$$u_n^{1/n} \sim \frac{a}{e} \exp\left(\frac{\log n}{\log \log n}\right) \frac{\log \log n}{\log n}. \quad (8.15)$$

This demonstrates why a theorem formulated only with $W(L(n))$ would be false for the stated general class.

9 A cascade of Borel-growth resonances

We now compute every term which survives at additive $o(1)$ precision in the logarithm of the ordinary Borel normalizer for (8.11). This is a finite transseries at each fixed γ , but its number of terms increases without bound as $\gamma \uparrow 1$.

For the moment keep $\gamma \in (0, 1)$ fixed, write

$$S = \log(at), \quad c_k(\gamma) = \frac{1}{k} \binom{k\gamma}{k-1}, \quad K_\gamma = \left\lfloor \frac{1}{1-\gamma} \right\rfloor. \quad (9.1)$$

Generalized binomial coefficients have their polynomial meaning here. The amplitude $a = c_1$ is not related to the coefficient $c_k(\gamma)$; the latter always carries its argument to avoid confusion.

Theorem 9.1 (All nonvanishing logarithmic growth terms). *For the stretched-logarithmic family and exponentially controlled positive amplitudes,*

$$\log \nu_1(t) = S + \sum_{k=1}^{K_\gamma} c_k(\gamma) S^{1-k(1-\gamma)} - \gamma \log S + o(1), \quad (9.2)$$

$$\log \mathcal{B}_1(t) \sim \frac{at}{S^\gamma} \exp\left(\sum_{k=1}^{K_\gamma} c_k(\gamma) S^{1-k(1-\gamma)}\right). \quad (9.3)$$

The theorem permits all the slope perturbations in (2.2).

Proof. Put $y = \log \nu_1(t)$. The defining equation for ν_1 and (8.11) give

$$y = S + y^\gamma - \gamma \log y + o(1). \quad (9.4)$$

Let Y be the unique large solution of $Y = S + Y^\gamma$. With $w = Y/S$ and $T = S^{\gamma-1}$, its equation is $w = 1 + Tw^\gamma$. The analytic implicit-function theorem applies near $(T, w) = (0, 1)$, and classical Lagrange inversion gives the convergent expansion

$$w = 1 + \sum_{k \geq 1} \frac{1}{k} \binom{k\gamma}{k-1} T^k. \quad (9.5)$$

Retaining terms through K_γ leaves an error $O(S^{1-(K_\gamma+1)(1-\gamma)}) = o(1)$ in Y .

We also have

$$y = Y - \gamma \log S + o(1). \quad (9.6)$$

To check this at the required additive precision, substitute $Y - \gamma \log S$ in $z - z^\gamma + \gamma \log z$. Using $Y/S \rightarrow 1$ and $S^{\gamma-1} \log S \rightarrow 0$, the residual from S is $o(1)$. The derivative of this function tends to one, so the residual produces an $o(1)$ error in its large inverse. This proves (9.6), and then (9.2). Exponentiate and use Theorem 2.5 to obtain (9.3). $\square$

Corollary 9.2 (Resonant constants). *At $\gamma = 1 - 1/m$, $m \geq 2$, the last term in (9.2) is the constant $1/m$. Hence*

$$\log \mathcal{B}_1(t) \sim e^{1/m} \frac{at}{S^{1-1/m}} \exp \left(\sum_{k=1}^{m-1} c_k (1 - 1/m) S^{1-k/m} \right). \quad (9.7)$$

Proof. At this parameter value $1 - m(1 - \gamma) = 0$ and $c_m(\gamma) = m^{-1} \binom{m-1}{m-1} = 1/m$. $\square$

For unit amplitudes, so that $S = \log t$, the first examples are

$$0 < \gamma < \frac{1}{2} : \quad \log \mathcal{B}_1(t) \sim t S^{-\gamma} e^{S^\gamma}, \quad (9.8)$$

$$\gamma = \frac{1}{2} : \quad \log \mathcal{B}_1(t) \sim e^{1/2} t S^{-1/2} e^{\sqrt{S}}, \quad (9.9)$$

$$\gamma = \frac{2}{3} : \quad \log \mathcal{B}_1(t) \sim e^{1/3} t S^{-2/3} \exp \left(S^{2/3} + \frac{2}{3} S^{1/3} \right), \quad (9.10)$$

$$\gamma = \frac{3}{4} : \quad \log \mathcal{B}_1(t) \sim e^{1/4} t S^{-3/4} \exp \left(S^{3/4} + \frac{3}{4} S^{1/2} + \frac{15}{32} S^{1/4} \right). \quad (9.11)$$

The term called a resonance here is a zero exponent in the logarithmic normalizer. It is not a collision of Borel singularities: the Borel transforms are entire. Omitting the constant term changes an asymptotic equivalent into an incorrect one even though it does not change the order or the infinite-type classification.

9.1 Every factorial normalization

The same argument, with $R = \sigma^{-1} \log(at)$, gives the following more general formula:

$$\log \mathcal{B}_\sigma(t) \sim \sigma(at)^{1/\sigma} R^{-\gamma/\sigma} \exp \left(\sum_{k=1}^{K_\gamma} \frac{c_k(\gamma)}{\sigma^k} R^{1-k(1-\gamma)} \right). \quad (9.12)$$

Indeed, $y = \log \nu_\sigma$ now obeys $y = R + \sigma^{-1} y^\gamma - (\gamma/\sigma) \log y + o(1)$, and its analytic core has $T = \sigma^{-1} R^{\gamma-1}$ in (9.5). At $\gamma = 1 - 1/m$, the corresponding constant factor is $\exp(1/(m\sigma^m))$. This formula uses $(n!)^\sigma$; for the $\Gamma(1 + \sigma n)$ normalization one must first make the argument change in (7.3).

9.2 Uniform transition through a resonance

The constant-factor jump in a fixed-parameter expansion has a smooth transition when the parameter is allowed to move. We state it for a fully specified family to make uniformity unambiguous:

$$\lambda_j(\gamma) = j \exp((\log(j+1))^\gamma), \quad c_j = 1, \quad 0 < \gamma < 1. \quad (9.13)$$

Write $\mathcal{B}_{1,\gamma}$ for its ordinary Borel transform.

Lemma 9.3 (Uniformity on compact parameter intervals). *For (9.13), the coefficient-root theorem and the Borel-growth theorem hold uniformly for γ in each compact subinterval $J \subset (0, 1)$. On J , the expansion*

$$b_\gamma(e^s) = s^\gamma - \gamma \log s + o(1) \quad (9.14)$$

also holds uniformly.

Proof. Let $\gamma_- \leq \gamma \leq \gamma_+$ describe J . The profile $L_\gamma(x) = \exp((\log(x+1))^\gamma)$ is admissible. For convexity, set $\ell = \log(x+1)$; then

$$(xL_\gamma(x))'' = \frac{\gamma L_\gamma(x) \ell^{\gamma-2}}{(x+1)^2} (\ell(x+2) + x(\gamma \ell^\gamma + \gamma - 1)) > 0.$$

The last sign follows from $\ell(x+2) - x > 0$, using $\log(1+x) \geq x/(1+x)$, and the remaining γ -terms are positive. Moreover

$$\sup_{\gamma \in J} \frac{xL'_\gamma(x)}{L_\gamma(x)} = O((\log x)^{\gamma_+-1}) \rightarrow 0,$$

and $\inf_{\gamma \in J} \log L_\gamma(x) \rightarrow \infty$. Thus $b_\gamma \rightarrow \infty$, $b_\gamma = o(\log x)$, and all the compact profile estimates in Lemma 5.1 are uniform. Its coercivity proof is uniform as well: the same eventual bound on the elasticity works for every $\gamma \in J$, and the bounded-index slopes are uniformly bounded. A contrary sequence $\gamma_n \in J$ would therefore contradict exactly that proof. This establishes the uniform root law.

In Lemma 7.1, the coefficient errors $\eta_n(\gamma)$ consequently tend to zero uniformly. The derivative bound for $\log f_\gamma$ and the small, central, and large index estimates are all uniform; the corresponding $N(t, \gamma)$ tend to infinity uniformly. This proves uniform Borel transfer. Finally, in Proposition 8.1, $h_\gamma(s) = s^\gamma + o(1)$ with the same uniform derivative estimates; $h'_\gamma \log h_\gamma = O(s^{\gamma_+-1} \log s) = o(1)$. The Taylor remainder is uniform, proving (9.14). $\square$

Theorem 9.4 (A double-scaling resonance profile). *Fix $m \geq 2$, put $S = \log t$, and let*

$$\gamma(t, \theta) = 1 - \frac{1}{m} + \frac{\theta}{\log S}. \quad (9.15)$$

For θ in every fixed compact real interval,

$$\frac{\log \mathcal{B}_{1, \gamma(t, \theta)}(t)}{tS^{-\gamma(t, \theta)} \exp\left(\sum_{k=1}^{m-1} c_k(\gamma(t, \theta)) S^{1-k(1-\gamma(t, \theta))}\right)} \rightarrow \exp\left(\frac{e^{m\theta}}{m}\right) \quad (9.16)$$

uniformly in θ .

Proof. For large t , all the parameters lie in a fixed compact subinterval of $(0, 1)$, so Lemma 9.3 applies. The analytic implicit-function expansion (9.5) has a uniform radius and bounded remainders on a sufficiently small closed parameter interval about $1 - 1/m$. Truncate it after the m th term. The error in Y is

$$O\left(S^{1-(m+1)(1-\gamma(t, \theta))}\right) = O\left(S^{-1/m} e^{(m+1)\theta}\right) = o(1)$$

uniformly for bounded θ . The logarithmic perturbation estimate is uniform too. The m th retained term is

$$c_m(\gamma(t, \theta)) S^{1-m(1-\gamma(t, \theta))} = c_m(\gamma(t, \theta)) e^{m\theta} \rightarrow \frac{e^{m\theta}}{m}.$$

Subtract the first $m - 1$ terms, exponentiate, and apply the uniform Borel growth theorem. This proves (9.16). $\square$

At $\theta = 0$ this gives the factor $e^{1/m}$. The leading terms in the denominator must be evaluated at the *moving* value of γ ; replacing them by their limiting parameter values changes unbounded terms. The theorem asserts uniformity for bounded θ , not a simultaneous limit with $|\theta| \rightarrow \infty$ or $m \rightarrow \infty$.

10 Exact checks and numerical diagnostics

The supplied program separates exact finite verification from floating-point asymptotic diagnostics. None of the latter is used as a premise of a theorem.

10.1 Independent rational coefficient computations

For rational inputs, coefficients are generated without a symbolic algebra system. If $E_j(q) = \exp(\lambda_j \widehat{U}(q)) = \sum_{r \geq 0} e_{j,r} q^r$, then

$$e_{j,0} = 1, \quad e_{j,m} = \frac{\lambda_j}{m} \sum_{r=1}^m r u_r e_{j,m-r}, \quad u_n = \sum_{j=1}^n c_j e_{j,n-j}. \quad (10.1)$$

This follows by differentiating E_j , and supplies a triangular recurrence independent of enumerating Lagrange compositions.

The recorded run uses $\lambda_j = jH_j$. It compares the recurrence with direct composition enumeration, checks convex composition bounds, checks the finite lower and upper inequalities using exact fractions, and repeats a coefficient comparison for the nonconstant amplitudes $c_j = (j+1)/2$. For the latter, $1 \leq c_j \leq 2^{j-1}$, so (2.3) holds with $a = 1$ and $D = \log 2$. Separately, the coefficients of $w = 1 + Tw^\gamma$ are checked through degree nine for $\gamma = 1 - 1/m$, $2 \leq m \leq 8$, including $c_m(\gamma) = 1/m$.

The recorded output reports **467 exact assertions passed**, with rational feedback coefficients computed through order 24. These are finite arithmetic checks, not a formal proof of the limit theorems.

10.2 Finite coefficient envelopes

The first table uses unit amplitudes and the harmonic slopes (8.9). Its lower and upper columns are the logarithms of the two sides of (3.4), divided by n . The scale b_n is computed with the smooth harmonic interpolation, and m_{low} maximizes the lower envelope.

n	lower $\log u_n/n$	b_n	upper $\log u_n/n$	scaled maximizer
100	1.383463	1.252376	1.999557	0.833379
1,000	1.618715	1.486732	2.270770	0.855436
10,000	1.783647	1.665473	2.433987	0.874808
100,000	1.915549	1.809832	2.559553	0.889537

The two envelopes need not lie on opposite sides of the asymptotic normalizer b_n : they enclose $\log u_n/n$, not its limit expression. In fact the lower column already exceeds b_n here. The convergence is very slow, and the upper estimate deliberately counts all compositions using a common maximal slope sum. At the displayed sizes the table does not identify the limiting constant by itself. These floating-point values are not outward-rounded interval certificates.

10.3 Resonant normalizer diagnostics

For the next table use the large- x profile $L(x) = \exp((\log x)^\gamma)$ and let $S = \log t$. Instead of forming enormous numbers t or $\nu_1(t)$, solve directly for $b = \log(\nu_1(t)/t)$:

$$b + \log b = (S + b - \log(1 + b))^\gamma. \quad (10.2)$$

At $\gamma = 1 - 1/m$, the displayed quantity is

$$\exp\left(b + \gamma \log S - \sum_{k=1}^{m-1} c_k(\gamma) S^{1-k/m}\right),$$

whose limit is $e^{1/m}$. The calculation uses 75 decimal working digits.

γ	$\log t$	normalizer ratio	limit $e^{1/m}$
1/2	10^4	1.6446049088	1.6487212707
1/2	10^8	1.6486602121	1.6487212707
1/2	10^{12}	1.6487206526	1.6487212707
2/3	10^4	0.9366736042	1.3956124251
2/3	10^8	1.3456993528	1.3956124251
2/3	10^{12}	1.3921133863	1.3956124251
3/4	10^4	0.4031221486	1.2840254167
3/4	10^8	1.0358703247	1.2840254167
3/4	10^{12}	1.2439118048	1.2840254167

These are diagnostics of the *implicit growth normalizer*, not numerical evaluations of $\mathcal{B}_1(t)$ or proofs that the same finite values approximate it closely. The JSON output also contains six moving-parameter normalizer checks for Theorem 9.4, computed at 100 decimal working digits; these have the same limitation. Even $S = 10^{12}$ leaves a visible subleading correction for $\gamma = 3/4$. Retaining the negative-power terms in (9.5) and the next moving-argument correction would substantially improve numerical convergence.

10.4 Reproduction

From the archive root, the commands are

```
python -m pip install -r requirements.txt
python code/verify.py
pdflatex -interaction=nonstopmode -halt-on-error near_linear_feedback.tex
pdflatex -interaction=nonstopmode -halt-on-error near_linear_feedback.tex
```

Exact arithmetic uses the standard Python library; NumPy and SciPy supply the large finite envelopes, and mpmath supplies high-precision normalizer calculations. The script writes both machine-readable output and the two tables included above. A third LaTeX pass may be used when section or table placement has changed.

11 Ten further research questions

11.1 A multiplicative coefficient equivalent

Determine the $\exp(o(n))$ factor in $u_n = (ae^{b_n})^n \exp(o(n))$, first for $\lambda_j = jH_j$ and unit amplitudes. The finite upper envelope ignores the exact distribution of slope sums across compositions. A solution will require finer entropy estimates and a saddle correction, not just a higher-order expansion of b_n . A useful intermediate target is the first sublinear term of $\log u_n$ with a proved error smaller than that term.

11.2 A local limit theorem for composition length

The rate function has $I''(1) = 1$, suggesting fluctuations of order $x_n/\sqrt{n}$ around a suitably corrected value of $n - K_n + 1$. That scale and a Gaussian law are conjectural here. Terms discarded as $o(n)$ can shift the centre by much more than the fluctuation width. Determine the correct centre and variance before asserting a central limit theorem, and then seek a uniform lattice local limit.

11.3 Condensation of the primitive actions

Does a typical positively weighted Lagrange composition have one exceptional part containing almost all its excess? The present length law neither proves nor disproves this. For $L(x) = \exp((\log x)^\gamma)$, splitting an action changes L by a relative amount of order $(\log x)^{\gamma-1}$ on fixed-ratio scales. This points to a possible sub- n rate for condensation. Establish that rate, identify competing entropy, and distinguish unique condensation from several macroscopic parts.

11.4 Minimal hypotheses and oscillatory slow variation

Remove the differentiability and convexity assumptions while retaining a correct normalizer. Convexity provides a sharp exceptional-part upper bound; without it, the relevant object may be an upper convex envelope of the discrete slopes rather than $jL(j)$ itself. Seek necessary and sufficient conditions under which that envelope preserves coefficient-root growth, and analyze slowly varying factors with long plateaux or oscillatory local elasticity.

11.5 Sparse actions and much smaller amplitudes

Allow $c_j = 0$, noninteger actions, or amplitudes with $\log c_j$ more negative than order j . The proof here uses the ability to fill almost every remaining unit of degree with action-one factors, and it needs at least exponentially sized exceptional-action amplitudes. Determine how support gaps and amplitude suppression alter the optimal length, the Lambert balance, and the rate function. A first concrete case is $c_j = \exp(-j \log \log(j + 2))$.

11.6 Signed feedback and vector systems

For signed or complex amplitudes the positive coefficient model is only an absolute majorant. Develop verifiable noncancellation conditions that recover lower growth bounds. For vector equations, determine whether a one-dimensional effective slope emerges from positive matrix feedback, or whether several interacting large actions produce a different variational problem. Any extension must state its infinite-tail conditions rather than invoke a finite-dimensional theorem informally.

11.7 Growth away from the positive Borel ray

Determine the directional growth of $\mathcal{B}_\sigma$ on nonpositive rays and in sectors. Positivity fixes maximum modulus but does not prevent cancellation on other rays. Compare any resulting Laplace sums with the left-sector analytic fixed points constructed in the repository feedback article. Neither an entire Borel transform nor coefficient Gevrey bounds alone proves that the two analytic constructions coincide.

Editorial note (ProveIt, 2026-09-29). Partly answered, for unit amplitudes, by two packages filed later in `Analysis/Transseries/docs/series-and-transseries/`. The near-linear class has $A = \limsup \lambda_j / (j \log j)^2 = 0$, so `Negative_Ray_Summation_Exponential_Feedback/` (its Theorems `thm:threshold` and `thm:borel`) gives fine (fixed-width half-strip) Borel–Laplace summability of $\hat{U}$ and of its inverse in direction π , an entire inverse Borel transform with exponential growth on a tube around the negative ray, and sums that satisfy the literal kernel near the negative axis. Since $jL(j) = O((j \log(j+1))^{1+1/k})$ for every $k > 1$, `Sharp_Negative_Direction_Summability_Exponential_Feedback/` (its Theorems `thm:headline` and `thm:ksum`) gives k -summability in direction π for every $k > 1$, with sum the canonical left-sectorial solution. Neither package treats the growth of $\mathcal{B}_\sigma$ in sectors for general σ or amplitudes $c_j \neq 1$, and neither cites this question.

11.8 Compatible acceleration and optimal truncation

Use the scale ν_1 to construct an acceleration, if possible, which respects multiplication and the nonlinear feedback equation after an explicit analytic interpretation of its countable kernel. The damping threshold in Proposition 7.3 is only a necessary guide. For a specified sectorial realization, also establish uniform remainder bounds and an actual optimal-truncation theorem. Coefficient-root growth alone does not certify errors of truncated analytic solutions.

Editorial note (ProveIt, 2026-09-29). The uniform-remainder part is answered for unit amplitudes by `Sharp_Negative_Direction_Summability_Exponential_Feedback/` (its Theorem `thm:headline`): the canonical left-sectorial solution has strong Gevrey- s remainder bounds on every proper left sector for every $s > 0$, because $jL(j) = O((j \log(j+1))^{s+1})$. Its optimized truncation errors (Theorem `thm:logheadline`) assume power growth j^p with $p > 1$ and do not cover this class. The acceleration and an actual optimal-truncation theorem remain open.

11.9 Inversion and parameter limits beyond fixed resonances

Determine the exact root-growth scale of the compositional inverse of $\hat{U}$. Equality of the least Gevrey index is too weak to decide whether ae^{b_n} survives reversion. Separately, extend Theorem 9.4 to an increasing resonance index m , or to the transition $\gamma \uparrow 1$ at a rate depending on t . That limit requires uniform control of a growing number of reversion terms and should connect to the genuinely superlinear regime.

Editorial note (ProveIt, 2026-09-29). `Sharp_Weighted_Type_Formal_Reversion/` proves a kernel-independent zero-loss theorem (its Theorem `thm:zeroloss`): for log-convex weights with $M_n^{1/n} \rightarrow \infty$ reversion preserves the weighted type. It would settle the root scale of the inverse here if the weight e^{nb_n} were shown to be eventually log-convex; neither package proves this. See the next note.

Editorial note (ProveIt, 2026-09-29). Log-convexity is not needed. The later package `Exact_Weighted_Type_Beyond_Log_Convexity/` (batch 52; not independently reviewed) shows that reversion preserves the type for every weight that is subexponentially equivalent to a supermultiplicative weight with roots tending to infinity (its

Theorem `thm:classificationintro`), and that a nonunit linear coefficient c_1 divides the inverse type by $|c_1|$ (its Corollary `cor:scalarscaling`). By Lemma 4.1, b is eventually nondecreasing with $b \rightarrow \infty$ and $b = o(\log x)$; so, after arbitrary positive values are given to b_m at the finitely many small m where it is undefined, $V_n = \exp(n \max_{m \leq n+1} b_m)$ has nondecreasing roots, is supermultiplicative with $V_n^{1/n} \rightarrow \infty$, and eventually $\log(V_n/e^{(n+1)b_{n+1}}) = -b_{n+1} = o(n)$. The weight e^{mb_m} is therefore admissible, $\widehat{U}$ has type c_1 for it by Theorem 2.3, and the inverse coefficients q_n satisfy $\limsup_n |q_n|^{1/n} e^{-b_n} = 1$. Whether $|q_n|^{1/n} e^{-b_n}$ converges remains open. This deduction is editorial; it is not stated in either article.

11.10 Formal proofs and certified asymptotic computation

A practical Lean sequence starts with the finite coefficient formula, the discrete convex composition bound, and the exact amplitude inequalities. The next stages are the compact profile lemma, its tail coercivity, and the maximum-term transfer. These are substantial but separable targets; no full transseries datatype is needed for the first stage. On the numerical side, implement directed interval versions of (3.4), certified roots for (2.4), and explicit moduli for the $o(1)$ terms. Keep finite interval enclosures, asymptotic theorems, and Lean certificates as distinct output types.

12 Conclusions

The near-linear regime is not a degenerate version of the superlinear Gevrey calculation. It has its own self-consistent Lambert scale:

$$b_n e^{b_n} = L\left(\frac{n}{1+b_n}\right), \quad u_n^{1/n} \sim a e^{b_n}.$$

The coefficient theorem tolerates substantial amplitude variability and bounded linear slope errors. At the same precision it provides a universal large-deviation law for Lagrange composition length, with a rate function independent of the slowly varying profile.

The scale transfers to entire Borel growth through one more nonlinear inversion. That transfer distinguishes zero Gevrey type at every positive order from positive-direction summability: all the normalized transforms are entire but too large for their standard Laplace kernels. For stretched-logarithmic amplification the inversion yields a finite, explicit power–logarithmic expansion at each parameter, an infinite sequence of resonant constants, and a uniform transition profile through each fixed resonance. A faster slowly varying example shows why the moving argument in the initial Lambert equation is indispensable.

These conclusions answer the inspected repository’s near-linear question at coefficient-root and transform-growth precision under the stated hypotheses. A full multiplicative coefficient equivalent, general signed or sparse systems, and a compatible acceleration theory remain separate research tasks rather than consequences asserted without proof.

A Source snapshot and claim boundaries

A.1 Pinned repository source

The initial survey used the repository commit

a795fcffa4b3ece22761d81fbed3c43be8b81795.

The unchanged predecessor was rechecked at commit

04473354a0f3edff2d6c365caf26170ca6b88735.

The initial transseries project tree was

e37727630fd519201ceb2b2f24fdc71b238873d9.

The direct predecessor is the file `article.tex` in

`Analysis/Transseries/docs/series-and-transseries/
Exponential_Feedback_Regularity_Classification/`

Its blob hash is

c6a83d71973a12fe7ed970d2abc56e2b49e6808f.

Editorial note (ProveIt, 2026-09-29). The delivered text called the first identifier above a “repository root tree”. It is a commit (the batch-45 catalogue commit; see `docs/incoming/README.md`), and the wording was corrected. The “transseries project tree” identifier is indeed a tree.

The contiguous source range 1400–1605 contains the further-research section, including the near-linear, amplitude, and summation questions. The source range 350–530 contains the positive coefficient formula, its proof, and the convergence criterion. The group README initially recorded six unmerged arrivals of 29 September 2026 and recorded twelve at the final recheck, after a second intake. Both versions describe the arrivals as lacking claim-by-claim review and Lean formalization. We therefore reprove the needed finite algebra rather than treating the predecessor as a formally verified dependency.

A.2 Scope of the repository comparison

The review used the current transseries project README, the group README and its descriptions of the initial six arrivals, the subsequent intake descriptions at the final recheck, and the relevant contiguous parts of the feedback article. It was not a line-by-line audit of the entire canonical and companion volumes or every repository file. No repository branch was changed. The proposed placement of this package is a new research directory beside the predecessor, pending mathematical review and normal repository intake.

A.3 Status matrix

Item	Status and boundary
Formal coefficient formula	Classical tool; reproved with amplitudes and checked at finite orders.
Coefficient-root equivalent	Proved under admissibility, positivity, and exponential amplitude bounds; not a full coefficient equivalent.
Length large deviations	Proved with speed n ; does not identify a largest-action or Gaussian law.
Entire Borel growth	Proved on the positive ray and hence for maximum modulus; not a directional classification.
Positive summation obstruction	Proved for every standard finite Gevrey order; not an obstruction to all generalized summation.
Resonance cascade and layer	Proved for the stated families; fixed m , compact moving parameters.
Computation	467 exact finite assertions plus nonrigorous floating-point diagnostics; no new Lean proof.
Novelty	Extension of an explicit inspected repository question; external publication priority not exhaustively established.

B A proof-audit guide

The central technical point is Lemma 5.1. Its compact identity is exact before taking limits, and its coercivity argument is needed to discard the exponentially many but poorly weighted compositions. Counting all compositions by 2^n before localization would destroy the constant at coefficient-root precision. The amplitude hypothesis is useful precisely because its logarithmic cost is at most $D(n - K_n)$, which is $o(n)$ in the localized regime.

The next sensitive step is the transfer to Borel growth. An error $\exp(o(n))$ in coefficients is compatible with an equivalent for $\log \mathcal{B}_\sigma(t)$, but not with an equivalent for $\mathcal{B}_\sigma(t)$ itself. All the displays in Section 9 deliberately concern the logarithm of the Borel transform. Exponentiating those displays a second time without a stronger error estimate would be invalid.

Finally, when a reversion term has exponent zero it must be retained before exponentiating the logarithmic normalizer. This is the source of the constants in Corollary 9.2, and it is why the index K_γ includes the equality case. The moving-parameter theorem requires uniform estimates from the coefficient theorem onward; its proof does not merely insert a moving parameter into a pointwise asymptotic statement.

References

- [1] Vladimir Reshetnikov, ProveIt research repository. *A Sharp Regularity Classification for Countable Exponential-Feedback Transseries*. Research article filed 29 September 2026; source snapshot and precise path in Appendix A. [Pinned TeX source](#). The repository owner is given for attribution of the source location; no independent authorship or peer-review claim is intended.
- [2] Ira M. Gessel. Lagrange inversion. *Journal of Combinatorial Theory, Series A* **144** (2016), 212–249. doi:10.1016/j.jcta.2016.06.018; arXiv:1609.05988.
- [3] Sabine Jansen, Tobias Kuna, and Dimitrios Tsagkarogiannis. Lagrange inversion and combinatorial species with uncountable color palette. *Annales Henri Poincaré* (2021). doi:10.1007/s00023-020-01013-0; arXiv:2008.10862.
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